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CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS **DATE: 08/07/2026**

EXPERIMENT 1 -- DDL Commands (CREATE, ALTER, DROP, TRUNCATE, RENAME)

AIM:

To create and manage the database structure for a **Flight Reservation System** using Data Definition Language (DDL) commands such as **CREATE, ALTER, DROP, TRUNCATE, and RENAME**.

PROCEDURE:

1. Open MySQL and create the database for the Flight Reservation System.
2. Use the CREATE command to create required tables such as Flight, Passenger, and Booking with suitable attributes and data types.
3. Use the ALTER command to add, modify, or remove columns from an existing table.
4. Use the RENAME command to change the name of an existing table.
5. Use the TRUNCATE command to remove all records from a table while keeping its structure.
6. Use the DROP command to permanently delete an unwanted table from the database.
7. Use DESC table_name; or SHOW TABLES; after the operations to verify the changes.
8. Execute each command and observe the output to ensure that all DDL operations are performed successfully.

PROGRAM:

1. CREATE DATABASE

CREATE DATABASE Flight_Reservation;

Then select the database:

USE Flight_Reservation;

OUTPUT:

Output

Action Output

	#	Time	Action	Message	Duration / Fetch
✔	93	10:40:55	CREATE DATABASE flight_booking__system	1 row(s) affected	0.015 sec
✔	94	10:40:55	USE flight_booking__system	0 row(s) affected	0.000 sec

2. CREATE TABLE

```
CREATE TABLE Flight (
Flight_ID INT PRIMARY KEY,
Flight_Name VARCHAR(50),
Source VARCHAR(50),
Destination VARCHAR(50),
Departure_Time TIME,
Price DECIMAL(10,2)
);
DESC Flight;
```

OUTPUT:

4	11:19:49	USE Flight_Reservation	0 row(s) affected	0.000 sec
5	11:20:19	CREATE TABLE Flight (Flight_ID INT PRIMARY KEY, Flight_Name VARCHAR(50), Source VARCHAR(50), Dest...	0 row(s) affected	0.063 sec

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	
Flight_Name	varchar(50)	YES		NULL	
Source	varchar(50)	YES		NULL	
Destination	varchar(50)	YES		NULL	
Departure_Time	time	YES		NULL	
Price	decimal(10,2)	YES		NULL	

3.ALTER – ADD COLUMN

Add a new column called Available_Seats:

```
ALTER TABLE Flight
ADD Available_Seats INT;
DESC Flight;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	
Flight_Name	varchar(50)	YES		NULL	
Source	varchar(50)	YES		NULL	
Destination	varchar(50)	YES		NULL	
Departure_Time	time	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Available_Seats	int	YES		NULL	

4. ALTER – MODIFY COLUMN

Change the size of Flight_Name:

```
ALTER TABLE Flight
MODIFY Flight_Name VARCHAR(100);
DESC Flight;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	
Flight_Name	varchar(100)	YES		NULL	
Source	varchar(50)	YES		NULL	
Destination	varchar(50)	YES		NULL	
Departure_Time	time	YES		NULL	
Price	decimal(10,2)	YES		NULL	
Available_Seats	int	YES		NULL	

5. ALTER – DROP COLUMN

Remove the Available_Seats column:

```
ALTER TABLE Flight
DROP COLUMN Available_Seats;
DESC Flight;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

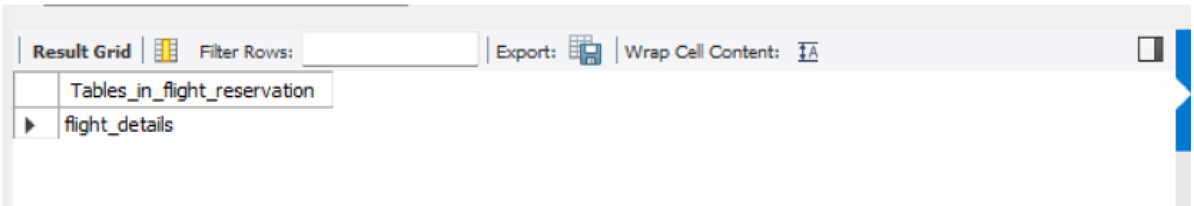
Field	Type	Null	Key	Default	Extra
Flight_ID	int	NO	PRI	NULL	
Flight_Name	varchar(100)	YES		NULL	
Source	varchar(50)	YES		NULL	
Destination	varchar(50)	YES		NULL	
Departure_Time	time	YES		NULL	
Price	decimal(10,2)	YES		NULL	

6. RENAME TABLE

Rename Flight to Flight_Details:

```
RENAME TABLE Flight TO Flight_Details;
SHOW TABLES;
```

OUTPUT:



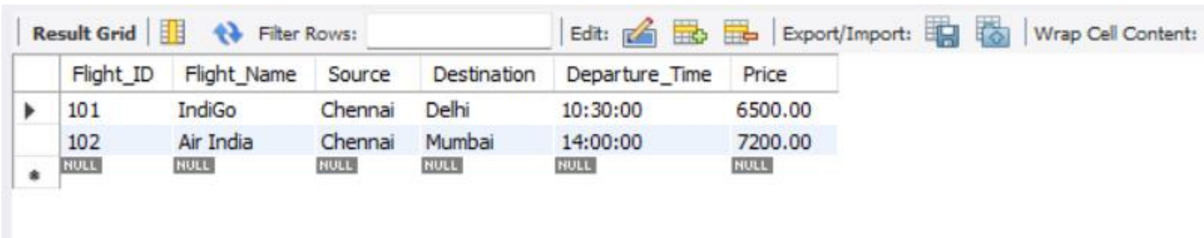
	Tables_in_flight_reservation
▶	flight_details

7. INSERT SAMPLE DATA

You need some data before testing TRUNCATE.

```
INSERT INTO Flight_Details
VALUES
(101, 'IndiGo', 'Chennai', 'Delhi', '10:30:00', 6500.00),
(102, 'Air India', 'Chennai', 'Mumbai', '14:00:00', 7200.00);
SELECT * FROM Flight_Details;
```

OUTPUT:



	Flight_ID	Flight_Name	Source	Destination	Departure_Time	Price
▶	101	IndiGo	Chennai	Delhi	10:30:00	6500.00
	102	Air India	Chennai	Mumbai	14:00:00	7200.00
*	NULL	NULL	NULL	NULL	NULL	NULL

8. TRUNCATE TABLE

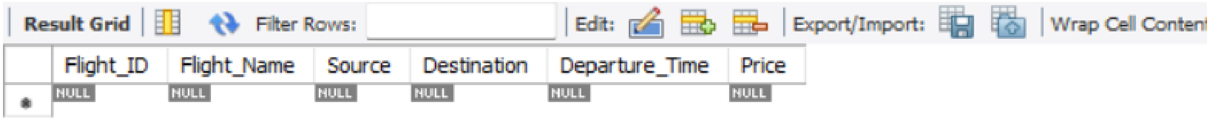
TRUNCATE TABLE Flight_Details;

Check:

SELECT * FROM Flight_Details;

The table will be empty, but the table still exists.

OUTPUT:

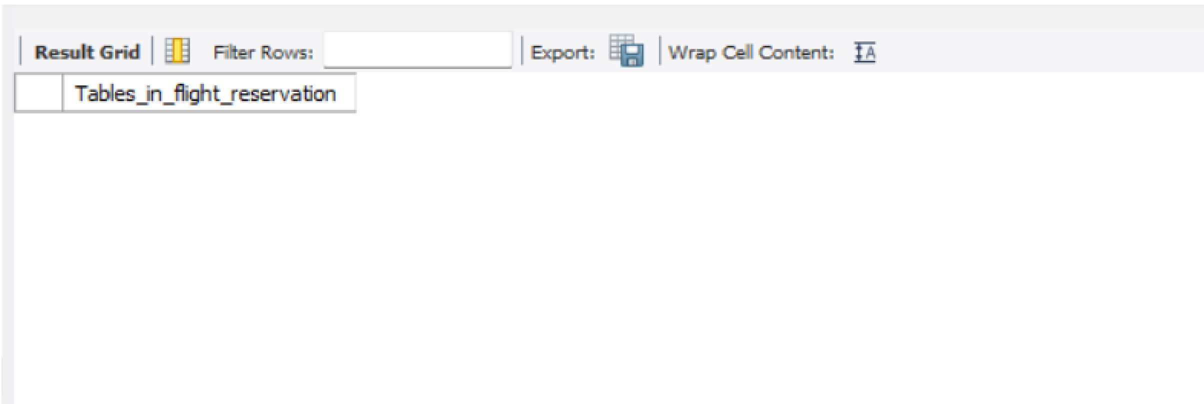


9. DROP TABLE

It completely removes the table.

DROP TABLE Flight_Details;
SHOW TABLES;

OUTPUT:



RESULT:

Thus the DDL commands executed successfully.